Equity and Inclusion

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Summary

The overall picture for equity and inclusion is broadly positive, though with important caveats. A substantial majority of FGD groups reported that the benefits of CFM are being shared equitably and that women benefit from CFM without disproportionately suffering opportunity costs. Notably, a majority of community female FGD groups themselves agreed that women have benefited from CFM. Carbon benefit-sharing agreements between carbon companies and communities are dominated by revenue sharing, while employment and livelihood support components are less commonly included. Responsibility for carbon income sharing decisions within communities was roughly equally split between instances where the CFMG executive makes the decisions and those where the whole CFMG is involved. Importantly, community female and youth FGD groups were significantly more likely than executive and male FGD groups to report inequitable benefit sharing, inadequate participation of women in decision making, and a lack of transparency in benefit sharing. In some CFMGs, community youth groups also identified issues with elite capture and marginalisation of youth. This common divergence in perspective between community members and CFMG executives is a concerning pattern suggesting that a minority of CFMGs have real equity problems that are not visible from the executive's perspective. Poor transparency in benefit sharing and the non-existence of grievance and redress mechanisms were also noted for many CFMGs. Without functioning formal channels through which equity concerns can be raised and addressed, inequitable outcomes risk going undetected and uncorrected. Strengthening these governance mechanisms is essential if CFM is to deliver equitable outcomes for all community members, including women, youth and other vulnerable groups.

Introduction

In this chapter, we consider the equitable sharing of benefits arising from community forest management (CFM) and what the impacts of CFM are on vulnerable groups.

Benefit sharing

We asked FGD groups whether the benefits of CFM were equitably distributed (Figure 1). On the one hand, a substantial majority of responses across the FGD groups asserted that benefits were being shared equitably. However, the proportion of community female and community youth groups that felt that benefits were not being shared equitably was significantly higher than for the CFMG executive and community male FGD groups. Moreover, even among the CFMG executive and community male groups, there were some that agreeded that benefits were not being shared equitably.

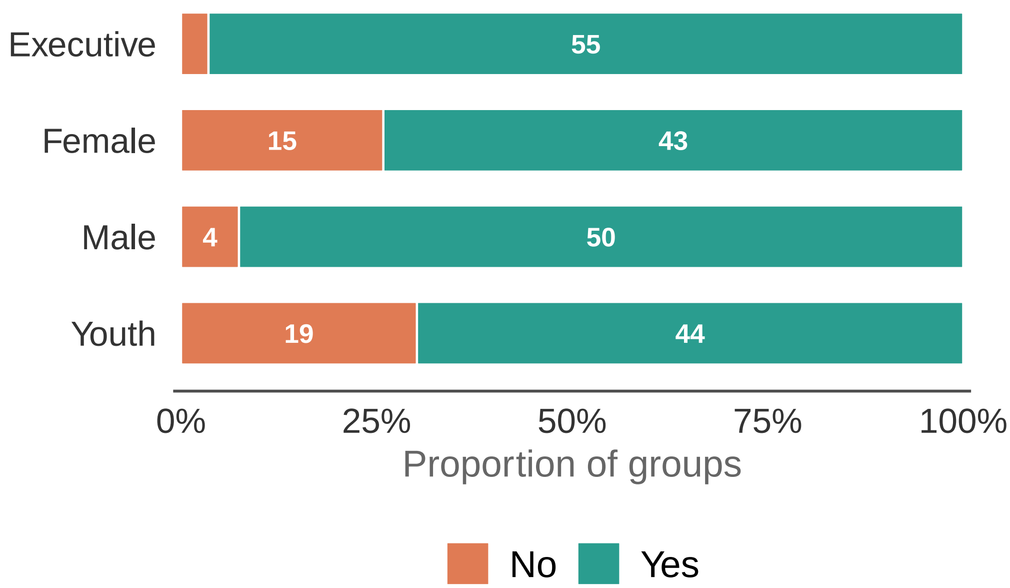

Figure 1: Are the benefits of CFM are equitably shared? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

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Random effect variances not available. Returned R2 does not account for random effects.
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Table 1: Results of the model examining the effect of FGD type and governance index on whether communities report equitable benefit sharing.

Model results for equitable benefit sharing <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0; CF-within-province variance = 2.144. Marginal R<U+00B2> = 0.412; conditional R<U+00B2> = NA. n = 217.

Factor	Estimate	SE	*P* value
Intercept (Executive FGD)	2.933	1.072	0.0062
Female FGD	−2.794	0.941	0.0030
Male FGD	−0.203	1.070	0.8498
Youth FGD	−2.870	0.922	0.0019
Governance index	0.259	0.157	0.0975

Table 2: Pairwise comparisons among FGD types for equitable benefit sharing.

Pairwise comparisons <U+2014> FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	*P* value
Executive - Female	2.794	0.941	2.970	0.0158
Executive - Male	0.203	1.070	0.189	0.9976
Executive - Youth	2.870	0.922	3.112	0.0100
Female - Male	−2.591	0.919	−2.818	0.0249
Female - Youth	0.077	0.514	0.149	0.9988
Male - Youth	2.668	0.910	2.933	0.0177

Individual interviewees were also asked whether the benefits of CFM were equitably shared. Table 3 shows the percentage reporting this broken down by interviewee role. Again a substantial majority of respondents (83% - 92%) thought that benefits were being shared equitably, and the differences among respondents were not significant. However, there was still minority of respondents that did not think benefits were being equitably shared. This suggests that there are some CFs where the benefit sharing arrangements are not optimal or appropriate arrangements are in place but not being followed.

Table 3: Percentage of individual interviewees reporting that the benefits of CFM are equitably shared by interviewee role.

Equitable sharing of benefits, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses. A positive response indicates that benefits of CFM are equitably shared.

Role	n	Agree benefits are equitably shared

Community member	31	92%
CFMG executive member	63	86%
Traditional leader	75	83%
District Forestry Officer	56	90%
NGO representative	32	96%

Table 4: Results of the model examining the effect of interviewee role, gender and age on whether individuals report equitable benefit sharing.

Model results for equitable benefit sharing <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.619; CF-within-province variance = 0.325. Marginal R<U+00B2> = 0.096; conditional R<U+00B2> = 0.298. n = 220.

Factor	Estimate	SE	*P* value
Intercept (Community member)	3.449	1.489	0.0205
CFMG executive member	−0.753	0.919	0.4128
Traditional leader	−0.938	0.931	0.3135
District Forestry Officer	−0.205	0.954	0.8295
NGO representative	0.903	1.321	0.4940
Male	−0.265	0.650	0.6831
Age	−0.011	0.021	0.6099

Carbon income benefit sharing

Company-community arrangements

CFMG executive FGDs reported on the components were included in their carbon finance benefit-sharing arrangement with the carbon trading enterprise (Figure 2). Clearly the agreements are dominated by revenue sharing, with relatively few also containing agreements on employment, infrastructure development and livelihood support.

Within-community sharing

We also asked FGD groups about the carbon income sharing arrangements within the community. We first asked who is responsible for the decision-making (Figure 3) and then we asked on what principles the benefit sharing was based (Figure 4). As can be seen there is broad agreement among the FGD groups on who makes the decision about the carbon benefit sharing arrangements. In approximately half the CFs the CFMG executive is responsible for the decision, while in the other half the entire CFMG is responsible.

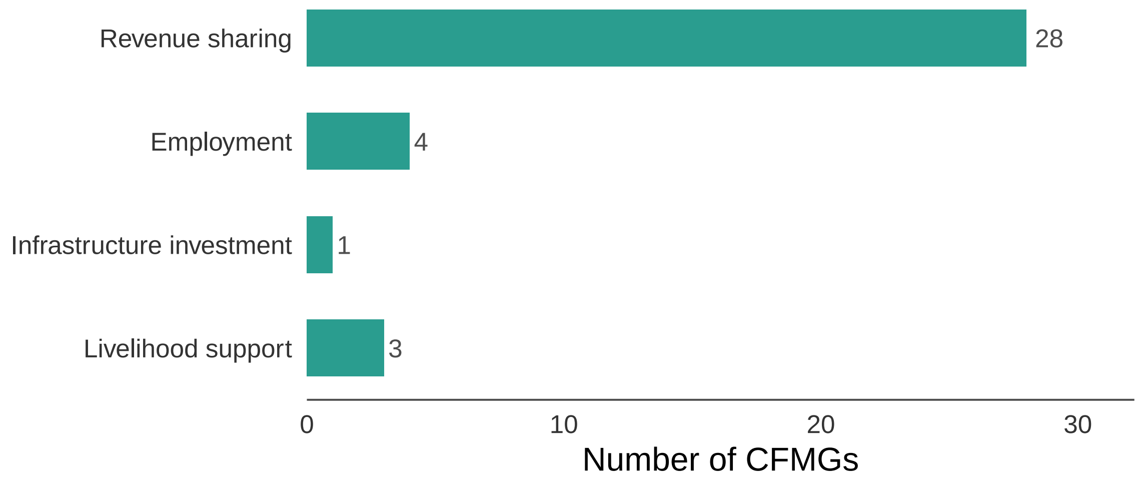

Figure 2: Components included in the carbon finance benefit-sharing arrangement between the carbon company and the community as reported by CFMG executive FGDs. The sample includes only those CFMGs that stated they had an agreement with a carbon trading company.

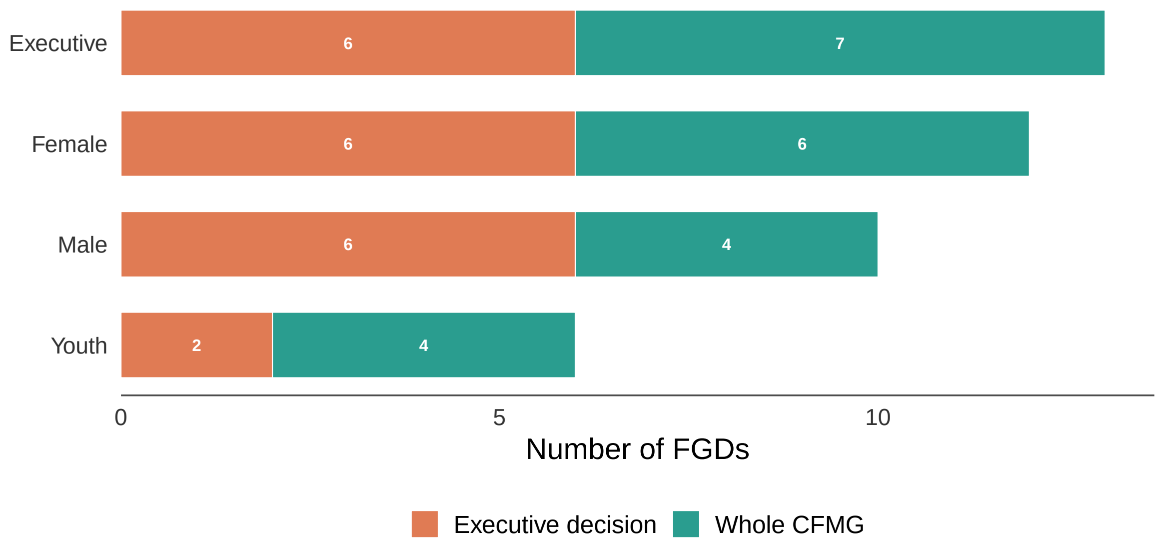

Figure 3: Who decides how carbon income is shared within the community by FGD group. The length of the segments is equal to the number of times that option was mentioned by the FGD group. Note: for 12 CFs the benefit-sharing arrangement is defined in the CFMG constitution.

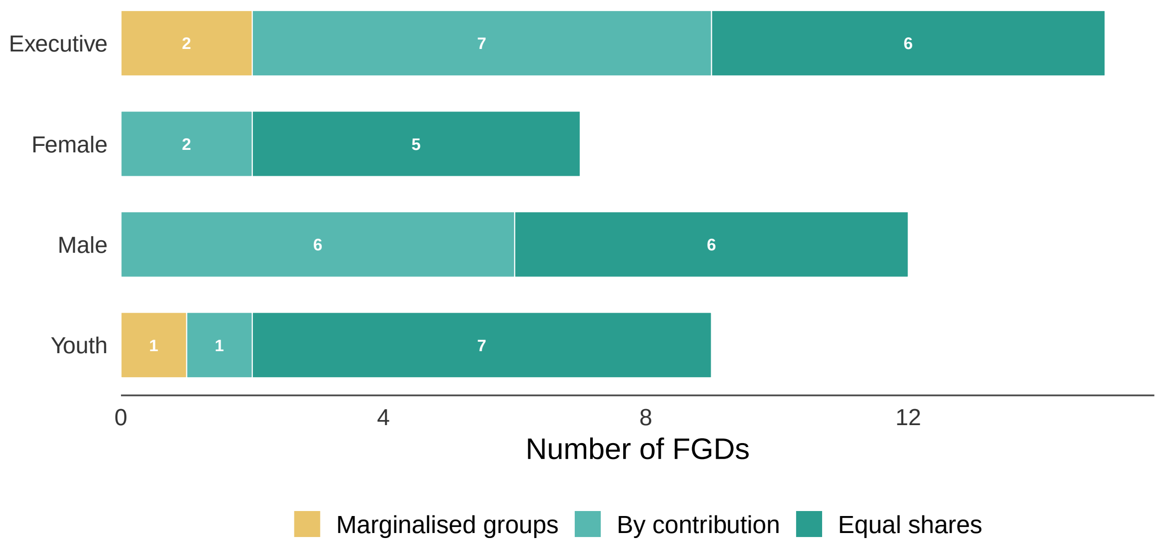

Figure 4: Principles governing carbon income sharing within the community by FGD group. The length of the segments is equal to the number of times that option was mentioned by the FGD group. Note: for 12 CFs the benefit-sharing arrangement is defined in the CFMG constitution.

According to the responses of the CFMG executives, seven CFs shared benefits according to members contribution to CFM while six shared them equally among the community households. Only two said that consideration of marginalised or vulnerable groups was a consideration. The other FGD groups were broadly similar in their responses and differences may be due to their knowledge of the arrangements, rather than actual differences in the arrangements.

External determination of benefit sharing

We also asked FGD groups whether stakeholders outside the community were also involved in determining the benefit-sharing arrangement (Figure 5).

Gender equity

We asked FGD groups whether women benefit from CFM (Figure 6).

Table 5: Results of the model examining the effect of FGD type and governance index on whether communities report that women benefit from CFM.

Model results for women benefit from CFM <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0; CF-within-province variance = 1.506. Marginal R<U+00B2> = 0.252; conditional R<U+00B2> = 0.487. n = 240.

Factor	Estimate	SE	*P* value
Intercept (Executive FGD)	3.317	1.000	0.0009
Female FGD	−2.453	0.861	0.0044
Male FGD	−0.955	0.924	0.3016
Youth FGD	−2.926	0.864	0.0007
Governance index	0.191	0.136	0.1605

Table 6: Pairwise comparisons among FGD types for whether women benefit from CFM.

Pairwise comparisons <U+2014> FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	*P* value
Executive - Female	2.453	0.861	2.847	0.0229
Executive - Male	0.955	0.924	1.033	0.7301
Executive - Youth	2.926	0.864	3.387	0.0039
Female - Male	−1.498	0.696	−2.151	0.1372
Female - Youth	0.473	0.500	0.946	0.7797
Male - Youth	1.971	0.698	2.825	0.0244

Individual interviewees were also asked whether women benefit from CFM. Table 7 shows the percentage reporting this broken down by interviewee role.

Table 7: Percentage of individual interviewees reporting that women benefit from CFM by interviewee role.

Women benefit from CFM, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses. A positive response indicates that women benefit from CFM.

Role	n	Agree women benefit from CFM
Community member	31	84%
CFMG executive member	63	95%
Traditional leader	75	79%
District Forestry Officer	56	94%

NGO representative	32	100%

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Random effect variances not available. Returned R2 does not account for random effects.
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Table 8: Results of the model examining the effect of interviewee role, gender and age on whether individuals report that women benefit from CFM.

Model results for women benefit from CFM <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0; CF-within-province variance = 7.489. Marginal R<U+00B2> = 0.95; conditional R<U+00B2> = NA. n = 239.

Factor	Estimate	SE	*P* value
Intercept (Community member)	1.820	2.180	0.4037
CFMG executive member	0.849	1.153	0.4613
Traditional leader	−1.249	1.088	0.2512
District Forestry Officer	1.661	1.141	0.1454
NGO representative	24.176	457.947	0.9579
Male	−0.520	0.941	0.5810
Age	0.048	0.032	0.1361

Women disproportionately suffer opportunity costs

We asked FGD groups whether women disproportionately suffer the opportunity costs of CFM (Figure 7).

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Random effect variances not available. Returned R2 does not account for random effects.
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Table 9: Results of the model examining the effect of FGD type and governance index on whether communities report that women disproportionately suffer opportunity costs of CFM.

Model results for women disproportionately suffering opportunity costs <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.126; CF-within-province variance = 0. Marginal R<U+00B2> = 0.394; conditional R<U+00B2> = NA. n = 268.

Factor	Estimate	SE	*P* value
Intercept (Executive FGD)	−5.132	1.154	0.0000
Female FGD	3.359	1.047	0.0013

Male FGD	0.202	1.423	0.8871
Youth FGD	2.280	1.085	0.0355
Governance index	0.159	0.105	0.1309

Table 10: Pairwise comparisons among FGD types for whether women disproportionately suffer opportunity costs of CFM.

Pairwise comparisons <U+2014> FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	*P* value
Executive - Female	−3.359	1.047	−3.209	0.0073
Executive - Male	−0.202	1.423	−0.142	0.9990
Executive - Youth	−2.280	1.085	−2.102	0.1524
Female - Male	3.157	1.050	3.008	0.0140
Female - Youth	1.079	0.496	2.173	0.1306
Male - Youth	−2.078	1.087	−1.913	0.2227

Individual interviewees were also asked whether women disproportionately suffer the opportunity costs of CFM. Table 11 shows the percentage reporting this broken down by interviewee role.

Table 11: Percentage of individual interviewees reporting that women disproportionately suffer the opportunity costs of CFM by interviewee role.

Women disproportionately suffer opportunity costs, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Agree women disproportionately suffer opportunity costs
Community member	31	29%
CFMG executive member	63	18%
Traditional leader	75	18%
District Forestry Officer	56	21%
NGO representative	32	7%

Table 12: Results of the model examining the effect of interviewee role, gender and age on whether individuals report that women disproportionately suffer opportunity costs of CFM.

Model results for women disproportionately suffering opportunity costs <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.293; CF-within-province variance = 0. Marginal R<U+00B2> = 0.09; conditional R<U+00B2> = 0.165. n = 239.

Factor	Estimate	SE	*P* value
Intercept (Community member)	−2.044	0.976	0.0363
CFMG executive member	−0.792	0.586	0.1765
Traditional leader	−0.946	0.584	0.1054
District Forestry Officer	−0.217	0.609	0.7217
NGO representative	−1.458	0.884	0.0992
Male	−0.549	0.431	0.2024
Age	0.031	0.015	0.0401

Women participate in decision making

We asked FGD groups whether women participate in CFM decision making (Figure 8).

Table 13: Results of the model examining the effect of FGD type and governance index on whether communities report that women participate in CFM decision making.

Model results for women participation in CFM decision making <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.491; CF-within-province variance = 1.446. Marginal R<U+00B2> = 0.356; conditional R<U+00B2> = 0.595. n = 267.

Factor	Estimate	SE	*P* value
Intercept (Executive FGD)	4.576	1.351	0.0007
Female FGD	−2.575	1.109	0.0202
Male FGD	−0.077	1.448	0.9575
Youth FGD	−3.957	1.146	0.0006
Governance index	0.118	0.148	0.4228

Table 14: Pairwise comparisons among FGD types for whether women participate in CFM decision making.

Pairwise comparisons <U+2014> FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	*P* value

Executive - Female	2.575	1.109	2.322	0.0930
Executive - Male	0.077	1.448	0.053	0.9999
Executive - Youth	3.957	1.146	3.453	0.0031
Female - Male	−2.498	1.114	−2.243	0.1118
Female - Youth	1.382	0.569	2.430	0.0716
Male - Youth	3.880	1.154	3.363	0.0043

Individual interviewees were also asked whether women participate in CFM decision making. Table 15 shows the percentage reporting this broken down by interviewee role.

Table 15: Percentage of individual interviewees reporting that women participate in CFM decision making by interviewee role.

Women participate in CFM decision making, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Agree women participate in decision making
Community member	31	100%
CFMG executive member	63	95%
Traditional leader	75	92%
District Forestry Officer	56	92%
NGO representative	32	97%

Table 16: Results of the model examining the effect of interviewee role, gender and age on whether individuals report that women participate in CFM decision making.

Model results for women participation in CFM decision making <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.891; CF-within-province variance = 0.359. Marginal R<U+00B2> = 0.902; conditional R<U+00B2> = 0.929. n = 252.

Factor	Estimate	SE	P value
Intercept (Community member)	21.616	14,068.386	0.9988
CFMG executive member	−19.157	14,068.386	0.9989
Traditional leader	−20.226	14,068.386	0.9989
District Forestry Officer	−19.524	14,068.386	0.9989
NGO representative	−18.832	14,068.386	0.9989
Male	0.246	0.749	0.7425
Age	0.019	0.030	0.5300

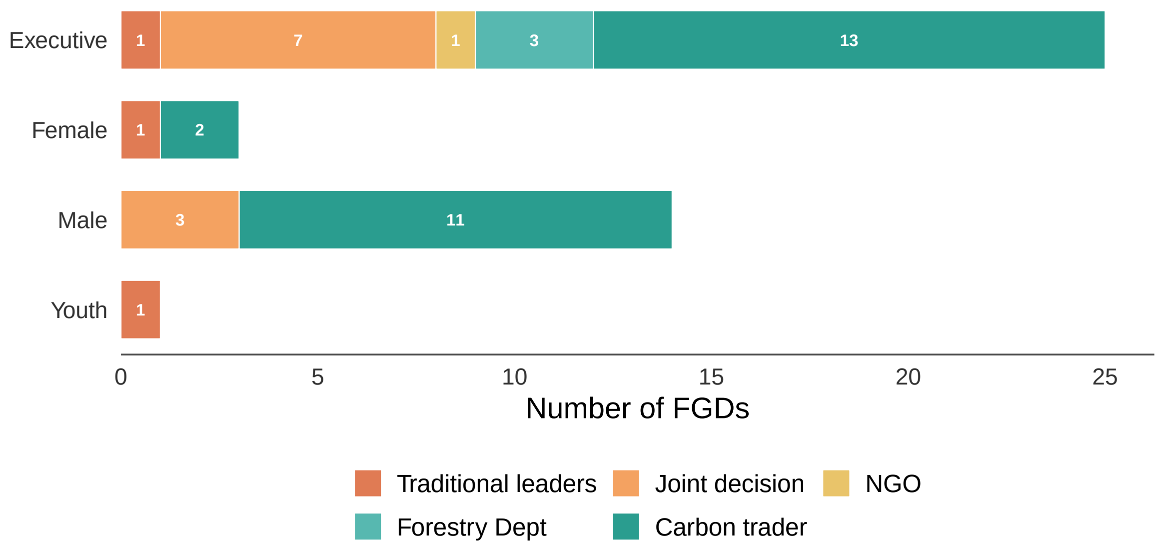

Figure 5: Who determines the benefit-sharing arrangement outside the community, by FGD group.

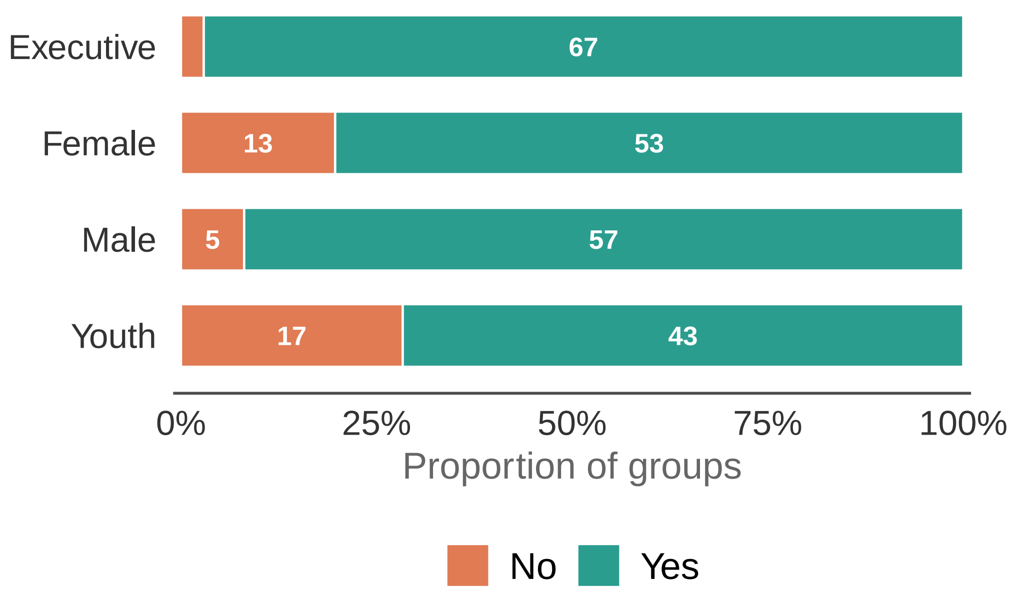

Figure 6: Do women benefit from CFM? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

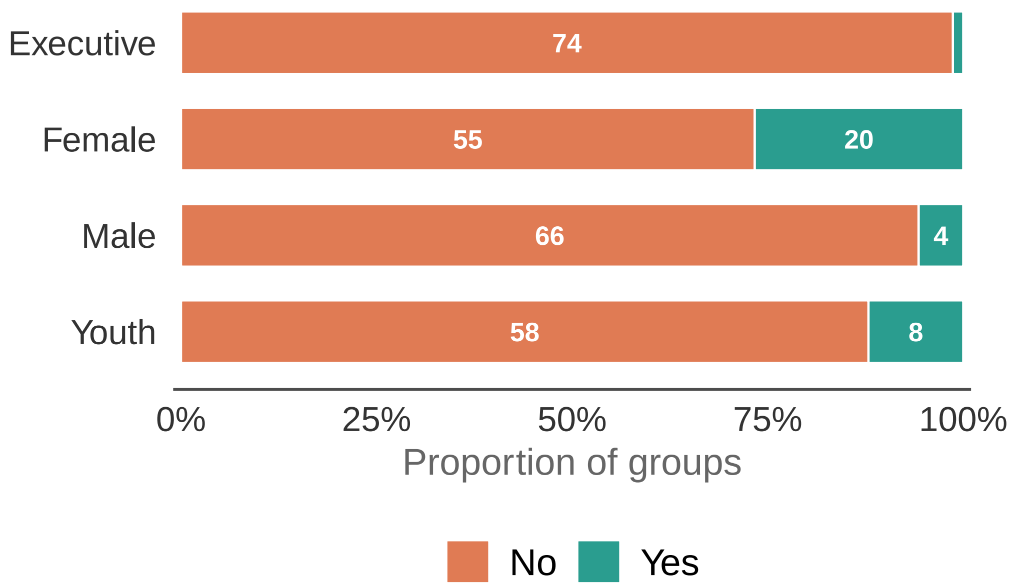

Figure 7: Do women disproportionately suffer the opportunity costs of CFM? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

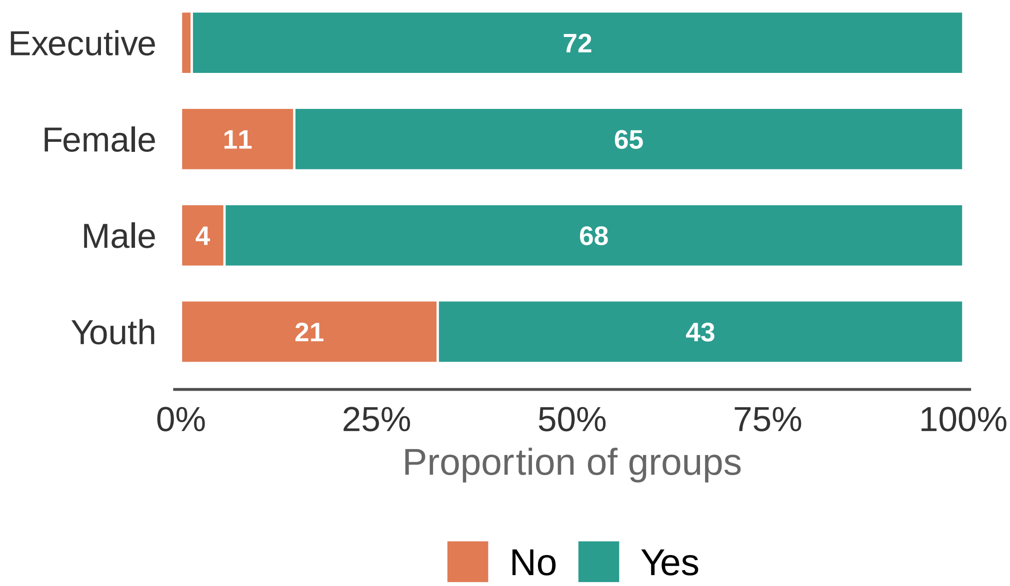

Figure 8: Do women participate in CFM decision making? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Transparency in benefit sharing

We asked FGD groups whether there is transparency in CFM benefit sharing (Figure 9).

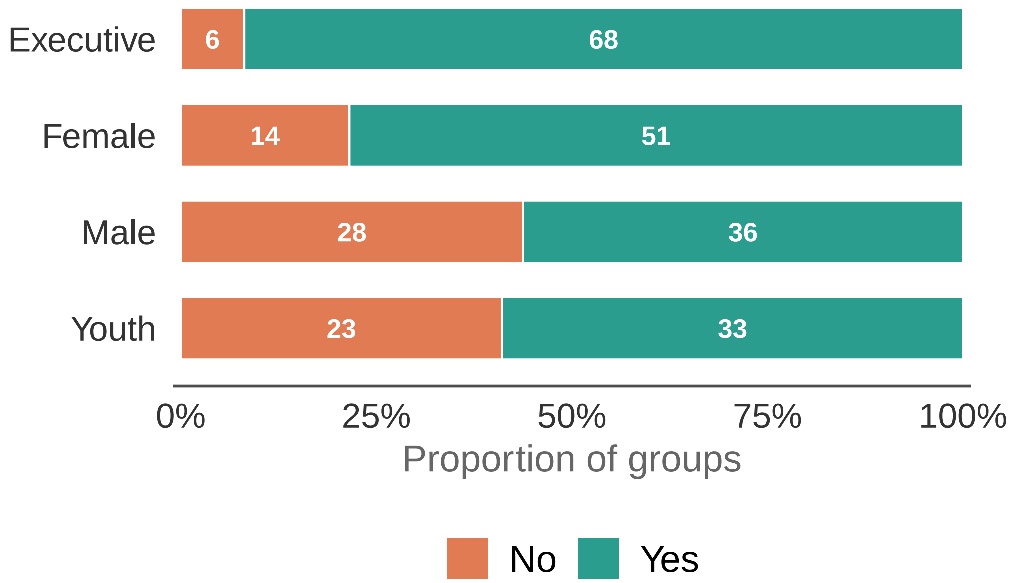

Figure 9: Is there transparency in CFM benefit sharing? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 17: Results of the model examining the effect of FGD type and governance index on whether communities report transparency in CFM benefit sharing.

Model results for transparency in CFM benefit sharing <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.142; CF-within-province variance = 0.306. Marginal R<U+00B2> = 0.278; conditional R<U+00B2> = 0.365. n = 242.

Factor	Estimate	SE	*P* value
Intercept (Executive FGD)	1.184	0.597	0.0472
Female FGD	−1.153	0.567	0.0420
Male FGD	−2.306	0.564	0.0000
Youth FGD	−2.023	0.559	0.0003
Governance index	0.340	0.104	0.0011

Table 18: Pairwise comparisons among FGD types for transparency in CFM benefit sharing.

Pairwise comparisons <U+2014> FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	P value
Executive - Female	1.153	0.567	2.033	0.1757
Executive - Male	2.306	0.564	4.085	0.0003
Executive - Youth	2.023	0.559	3.616	0.0017
Female - Male	1.153	0.472	2.440	0.0697
Female - Youth	0.869	0.472	1.841	0.2539
Male - Youth	−0.283	0.437	−0.648	0.9161

Individual interviewees were also asked whether there is transparency in CFM benefit sharing. Table 19 shows the percentage reporting this broken down by interviewee role.

Table 19: Percentage of individual interviewees reporting that there is transparency in CFM benefit sharing by interviewee role.

Transparency in CFM benefit sharing, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Agree there is transparency in benefit sharing
Community member	31	92%
CFMG executive member	63	93%
Traditional leader	75	76%
District Forestry Officer	56	79%
NGO representative	32	80%

Table 20: Results of the model examining the effect of interviewee role, gender and age on whether individuals report transparency in CFM benefit sharing.

Model results for transparency in CFM benefit sharing <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.44; CF-within-province variance = 0.465. Marginal R<U+00B2> = 0.123; conditional R<U+00B2> = 0.312. n = 237.

Factor	Estimate	SE	P value
Intercept (Community member)	3.305	1.416	0.0196
CFMG executive member	0.070	0.981	0.9431
Traditional leader	−1.529	0.869	0.0784

District Forestry Officer	−1.390	0.941	0.1397
NGO representative	−1.506	0.989	0.1280
Male	0.445	0.524	0.3961
Age	−0.016	0.019	0.3924

We examined which mechanisms are being used to achieve transparency in benefit sharing among the community female, male, and youth FGDs (Figure 10).

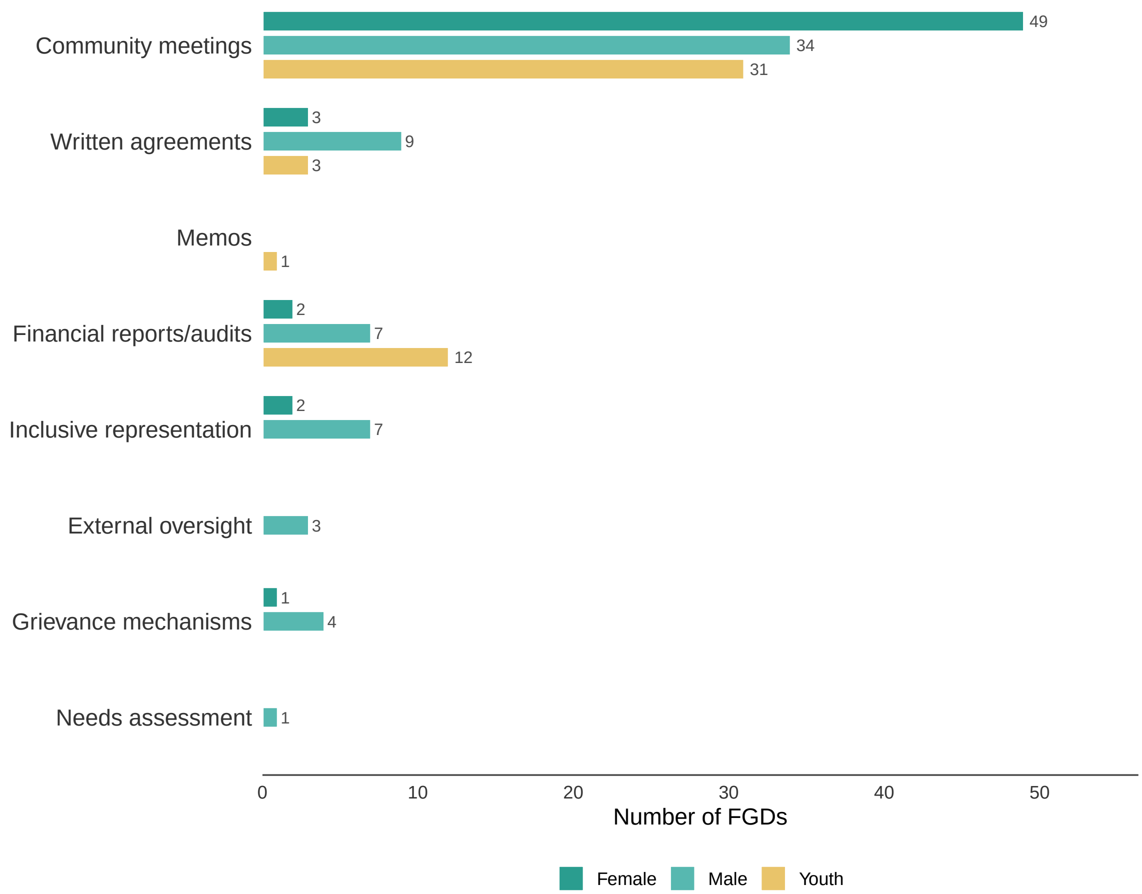

Figure 10: Mechanisms used to achieve transparency in CFM benefit sharing, as reported by community female, male, and youth FGDs. Bars show the number of FGDs reporting each mechanism.

Grievance and Redress

We asked FGD groups whether their CFMG has a grievance and redress mechanism in place (Figure 11).

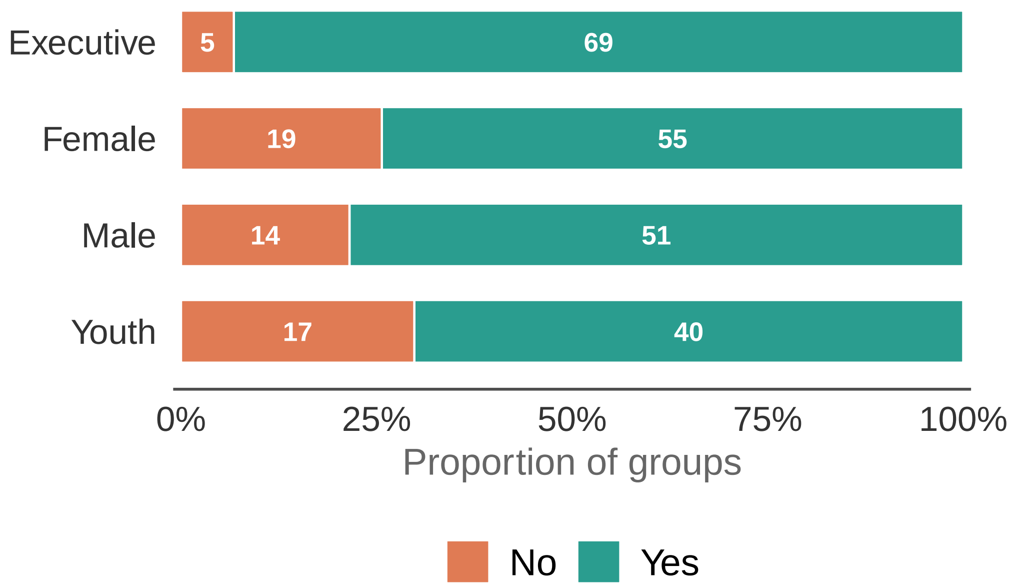

Figure 11: Does the CFMG have a grievance and redress mechanism? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

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Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   268.3    289.9   -128.1    256.3      264

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.1004  0.2075  0.4423  0.5798  0.7765

Random effects:
```

```
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.05611  0.2369
 province      (Intercept) 0.16831  0.4103
Number of obs: 270, groups:  province_cfmg, 75; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)            2.6416     0.5132   5.147 2.64e-07 ***
interview_typeFemale  -1.5702     0.5398  -2.909  0.00363 **
interview_typeMale    -1.3029     0.5598  -2.327  0.01994 *
interview_typeYouth   -1.7910     0.5551  -3.226  0.00125 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.809
intrvw_typM -0.784  0.723
intrvw_typY -0.793  0.730  0.704


 contrast           odds.ratio    SE  df null z.ratio p.value
 Executive / Female      4.807 2.595 Inf    1   2.909  0.0190
 Executive / Male        3.680 2.060 Inf    1   2.327  0.0918
 Executive / Youth       5.995 3.328 Inf    1   3.226  0.0069
 Female / Male           0.765 0.314 Inf    1  -0.652  0.9148
 Female / Youth          1.247 0.502 Inf    1   0.549  0.9470
 Male / Youth            1.629 0.699 Inf    1   1.137  0.6666

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
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Individual interviewees were also asked whether their CFMG has a grievance and redress mechanism. Table 21 shows the percentage reporting this broken down by interviewee role.

Table 21: Percentage of individual interviewees reporting that the CFMG has a grievance and redress mechanism by interviewee role.

Grievance and redress mechanism, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Grievance and redress mechanism
Community member	31	97%

CFMG executive member	63	90%
Traditional leader	75	86%
District Forestry Officer	56	77%
NGO representative	32	86%

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   199.2    230.5    -90.6    181.2      231

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.0701  0.2416  0.3224  0.4368  0.7224

Random effects:
 Groups        Name        Variance  Std.Dev.
 province_cfmg (Intercept) 1.417e-07 0.0003764
 province      (Intercept) 2.551e-01 0.5050848
Number of obs: 240, groups:  province_cfmg, 75; province, 8

Fixed effects:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                       3.249784   1.548580   2.099   0.0359 *
interviewee_rolecfmg_exec_member -1.205163   1.127959  -1.068   0.2853
interviewee_roletrad_leader      -1.578967   1.089285  -1.450   0.1472
interviewee_roledfo              -2.155322   1.122082  -1.921   0.0548 .
interviewee_roleNGO_rep          -1.588076   1.201137  -1.322   0.1861
interviewee_gendermale            0.790989   0.479019   1.651   0.0987 .
interviewee_age                  -0.009331   0.019952  -0.468   0.6400
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.687
intrvw_rlt_ -0.586  0.845
intrvw_rldf -0.786  0.858  0.854
```

```
intrvw_NGO_ -0.739  0.802  0.796  0.838
intrvw_gndr -0.171 -0.019 -0.082 -0.049    0.008
interview_g -0.714  0.125 -0.027  0.268    0.239 -0.036
optimizer (Nelder_Mead) convergence code: 0 (OK)
Model failed to converge with max|grad| = 0.00605841 (tol = 0.002, component 1)

 contrast                         odds.ratio    SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member        3.337 3.764 Inf    1   1.068  0.8228
 cfmg_member / trad_leader             4.850 5.283 Inf    1   1.450  0.5954
 cfmg_member / dfo                     8.631 9.684 Inf    1   1.921  0.3061
 cfmg_member / NGO_rep                 4.894 5.879 Inf    1   1.322  0.6774
 cfmg_exec_member / trad_leader        1.453 0.899 Inf    1   0.604  0.9745
 cfmg_exec_member / dfo                2.586 1.551 Inf    1   1.585  0.5073
 cfmg_exec_member / NGO_rep            1.467 1.078 Inf    1   0.521  0.9853
 trad_leader / dfo                     1.780 1.065 Inf    1   0.963  0.8717
 trad_leader / NGO_rep                 1.009 0.747 Inf    1   0.012  1.0000
 dfo / NGO_rep                         0.567 0.377 Inf    1  -0.853  0.9138

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

We also examined which mechanisms are being used to achieve grievance and redress among the community female, male, and youth FGDs (Figure 12).

Impact on poverty

We asked the community female, male, and youth FGDs to identify the ways in which CFM has had an impact on poverty in their communities (Figure 13).

Barriers to poorer community members

We asked the community female, male, and youth FGDs to identify the barriers that prevent poorer community members from benefiting from CFM (Figure 14).

Addressing challenges

We asked the community female, male, and youth FGDs to suggest solutions to address the barriers facing poorer community members in benefiting from CFM (Figure 15).

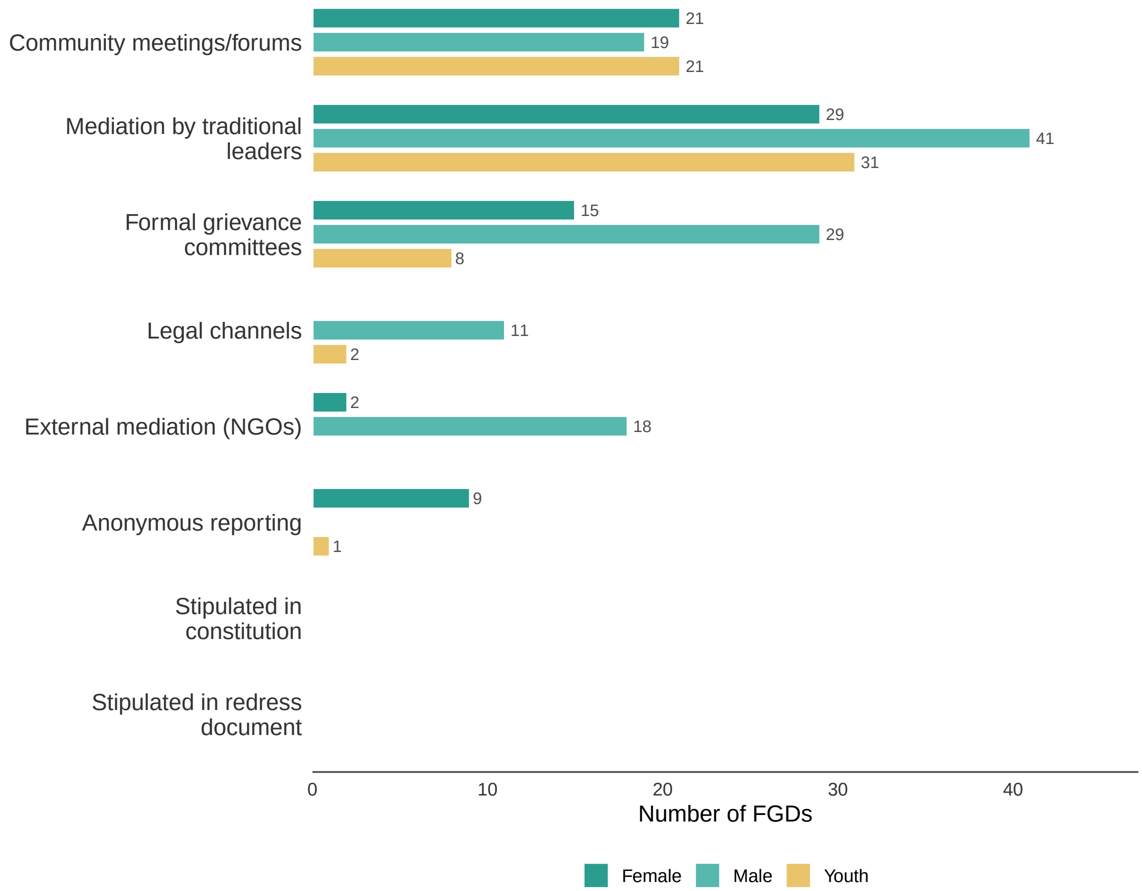

Figure 12: Mechanisms used for grievance and redress in CFMGs, as reported by community female, male, and youth FGDs. Bars show the number of FGDs reporting each mechanism.

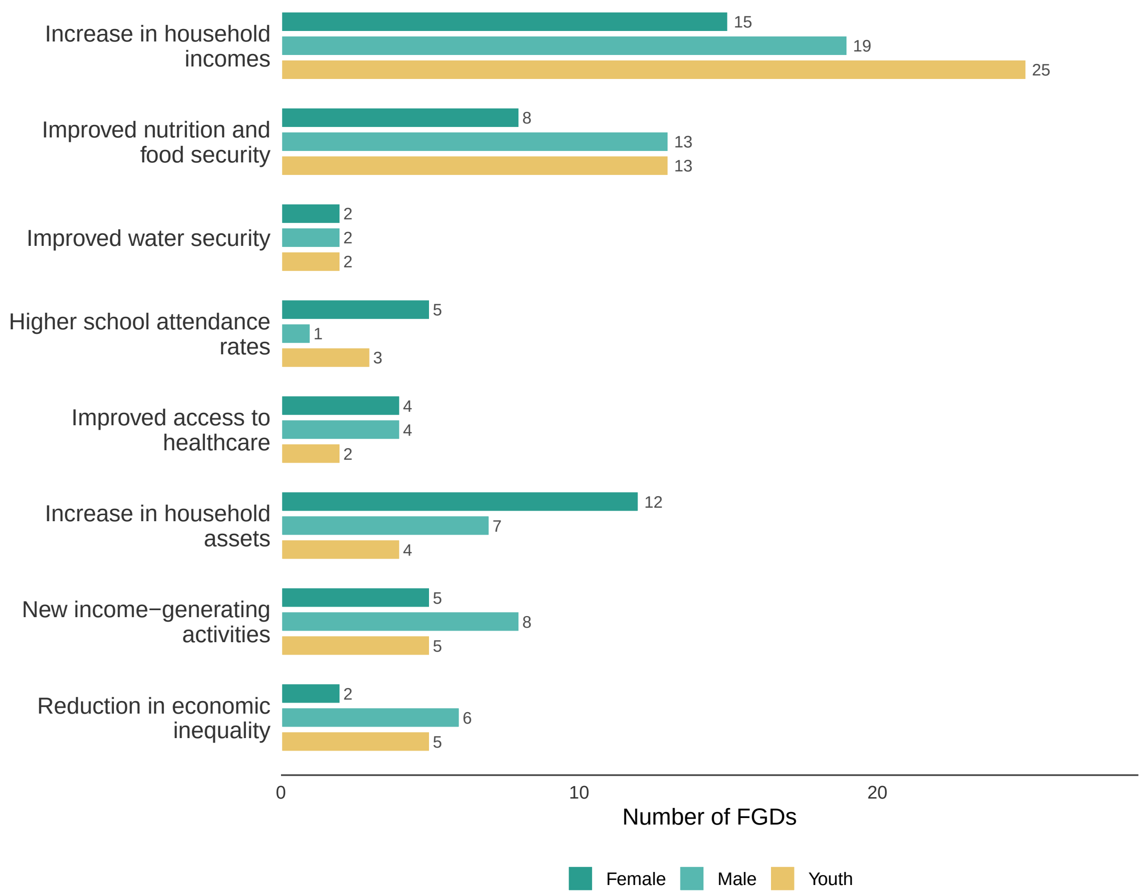

Figure 13: Reported impacts of CFM on poverty, as identified by community female, male, and youth FGDs. Bars show the number of FGDs reporting each impact.

Figure 14: Barriers preventing poorer community members from benefiting from CFM, as reported by community female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each barrier.

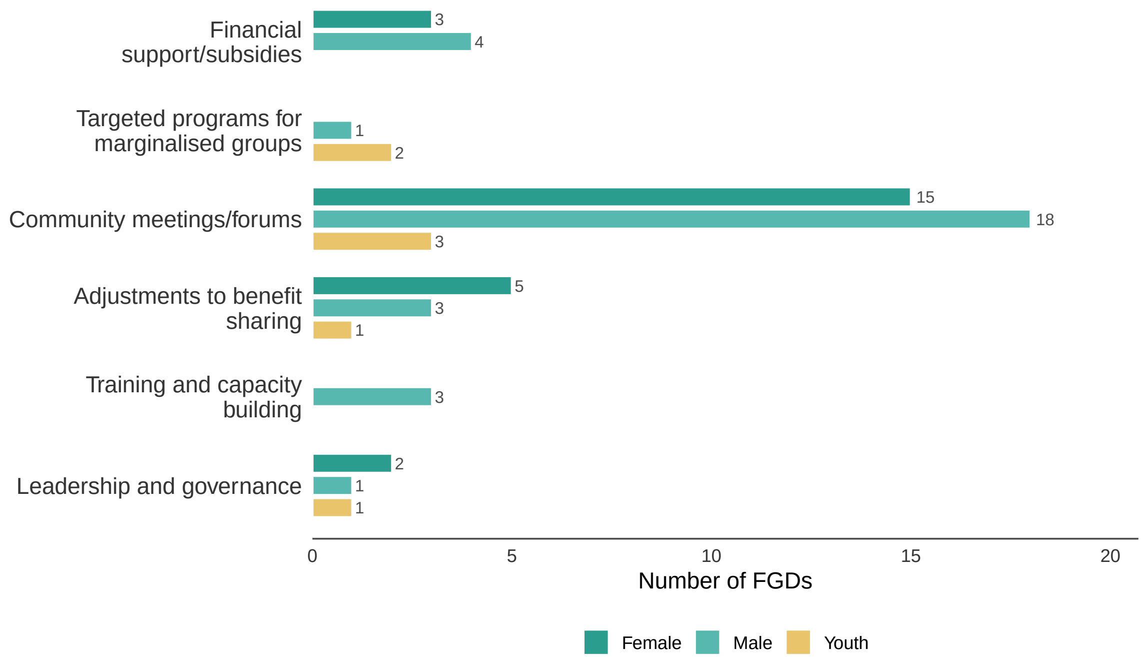

Figure 15: Suggested solutions to address barriers preventing poorer community members from benefiting from CFM, as reported by community female, male, and youth FGDs. Bars show the number of FGDs reporting each solution.

Equity and inclusion index

To summarise the overall equity and inclusion performance of each CFMG, we constructed an equity and inclusion score (0–5) by summing five binary indicators reported by FGD groups — equitable benefit sharing, women benefiting from CFM, women participating in decision making, transparency in benefit sharing, and existence of a grievance and redress mechanism — each scored as 1 (positive outcome reported) or 0 (no positive outcome). The distribution of equity and inclusion scores across CFMGs is shown in Figure 16. We then modelled this score using a linear mixed model with FGD type and the governance index as predictors, and random intercepts for province and CF within province, to test whether governance and FGD type were significant predictors of overall equity and inclusion performance (Table 22, Figure 17).

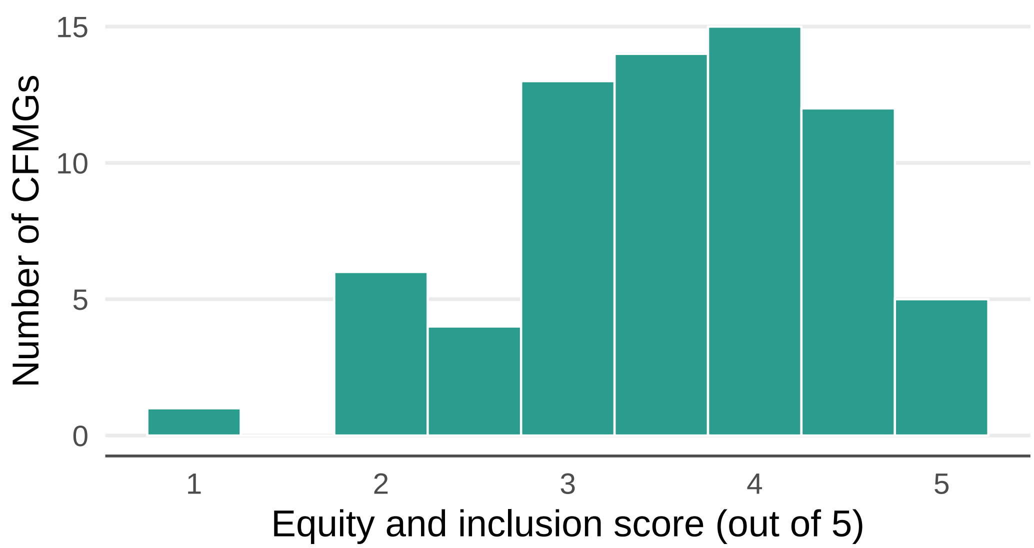

Figure 16: Distribution of mean equity and inclusion scores across CFMGs (0–5). Each bar represents the number of CFMGs with that mean score, averaged across all FGD types. The score sums five binary indicators: equitable benefit sharing, women benefiting from CFM, women participating in decision making, transparency in benefit sharing, and existence of a grievance and redress mechanism.

Table 22: Results of the model examining the effect of FGD type and governance index on equity and inclusion performance.

Model results for equity and inclusion performance <U+2014> FGD type and governance index

Linear mixed model with random intercepts for province and CF within province. Reference = Executive. Province variance = 0.095; CF-within-province variance = 0.233; residual variance = 1.355. Marginal R<U+00B2> = 0.19; conditional R<U+00B2> = 0.348. n = 273.

Factor	Estimate	SE	*P* value
Intercept	3.589	0.277	0.0000
Female	−0.873	0.195	0.0000
Male	−0.664	0.199	0.0010
Youth	−1.373	0.197	0.0000
Governance index	0.171	0.046	0.0004

Discussion

Through the FGDs and individual interviews, we investigated a wide range of equity and inclusion indicators, covering benefit sharing, gender equity, transparency, grievance and redress, and poverty impacts.

The most striking and consistent finding is a divergence in perspective by FGD group. Across virtually every indicator — equitable benefit sharing, women benefiting from CFM, women suffering disproportionate opportunity costs, women's participation in decision making, and transparency in benefit sharing — community female and youth groups were significantly more likely than CFMG executive and male groups to report negative outcomes. Given that CFMG executives and community males may typically be better informed about formal governance arrangements, it is possible that this divergence reflects an information asymmetry rather than disagreement about conditions. However, it may also reflect that the benefits and costs of CFM are being distributed unevenly, in ways that are visible to female and youth community members but not acknowledged — or perhaps not noticed — by those in the CFMG executives. It is also possible that these differences in perspective reflect elite capture or corruption.

Notwithstanding these these issues, the overall picture is more positive than might have been expected. A clear majority of all FGD groups — including community female groups — reported that benefits are being shared equitably and that women benefit from CFM. Most female groups also stated that women do not disproportionately suffer opportunity costs and that women participate in CFM decision making. This is somewhat surprising, given that natural resource management decisions in Zambia are traditionally the purview of male community members. It may be that the formal governance structures mandated by CFM — elections, executive positions with female representation, and documented benefit-sharing arrangements — are providing women with a degree of recognition and voice that they would not otherwise have had.

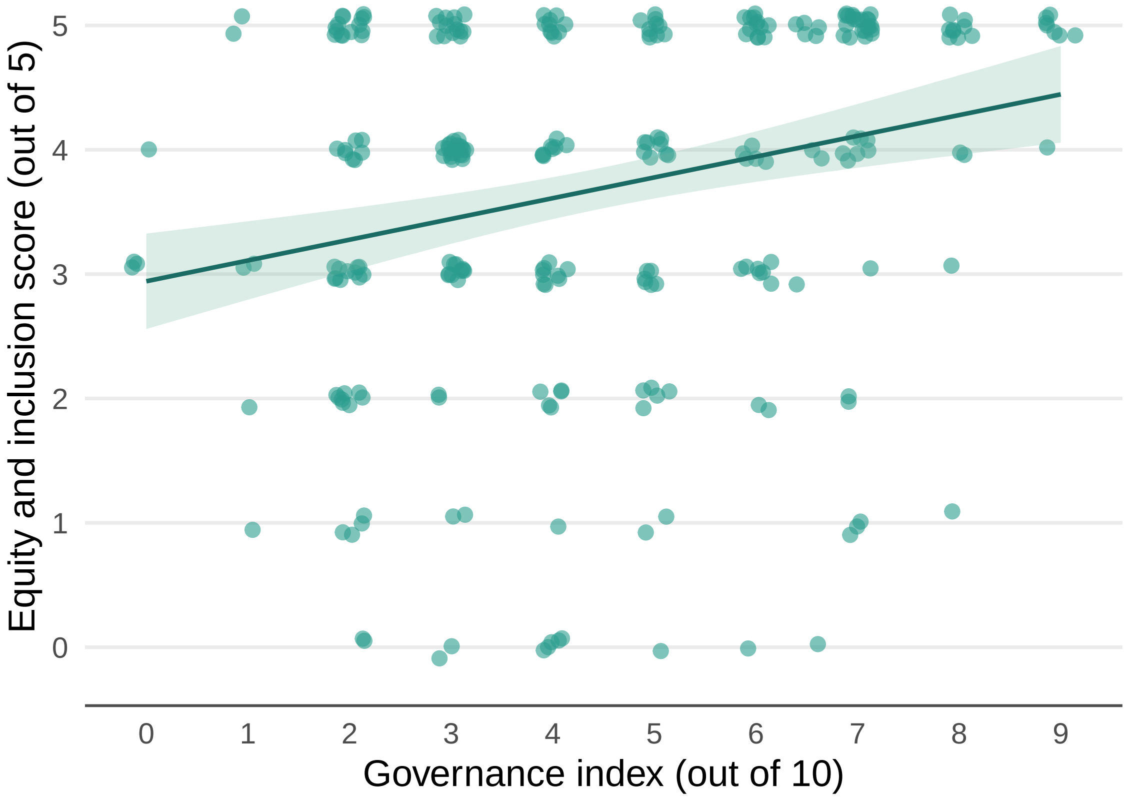

Figure 17: Relationship between the CFMG governance index and equity and inclusion score. Each point represents one FGD session. The line shows the fitted relationship from the linear mixed model, with the shaded band representing ± 1 SE.

Carbon benefit-sharing agreements between companies and communities are almost entirely structured around revenue sharing. Employment, infrastructure investment and livelihood support are less commonly included in these agreements. While revenue sharing provides a direct financial return, it means communities that have not yet received a carbon payment have little to show for CFM. Diversifying the components of company-community agreements could broaden both the distribution of benefits and community buy-in.

The reported poverty impacts of CFM — primarily increased household incomes and new income-generating activities — are broadly consistent with the economic transformation chapter findings. However, barriers to poorer community members benefiting remain, including limited access to resources, capital and markets. The solutions communities themselves proposed — financial support, targeted programmes for marginalised groups, and better governance — align closely with the governance and economic transformation recommendations identified elsewhere in this report.

In conclusion, the equity and inclusion outcomes of CFM in Zambia are broadly positive but uneven. The divergence between female and youth perspectives and those of CFMG executives is the most important finding of this chapter. It signals that equity problems in a subset of CFMGs are likely more widespread than executive-level data alone would suggest, and that strengthening transparency, grievance and redress mechanisms and community-wide engagement is essential to ensuring CFM benefits reach all community members equitably.